The Frohmanian Symplectic Tether $\mathfrak{T}_F$: Written Proofs Companion to the Lean 4 Formalization

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Abstract

This is a sibling paper to the Lean 4 repository `frohmanian-symplectic-tether-may-form`. Each result is given a written proof and cross-referenced to the Lean declaration that kernel-checks it. Result 1 is geometric: under three admissibility conditions, the symmetric correction $\mathfrak{T}_F[\kappa, w, u](F, G) = -(\kappa w)\langle \Pi_u F, \Pi_u G\rangle$ is unique up to scale, and Route A fixes the scale. Result 2 is about a *modified* system, the tethered vorticity equation $\partial_t\omega + (u\cdot\nabla)\omega = (\omega\cdot\nabla)u + \nu\Delta\omega - \kappa P(|\omega|^2\omega)$. For that system we prove the L^2 bound, the H^1 estimate, the log-loss comparison and global smoothness, *conditional on cited propositions that are listed explicitly.* **None of this resolves the Clay Millennium problem.** Plain Navier–Stokes is unchanged (Section 4).

1 Physical motivation

The starting picture is physical. The vortex-stretching term $(\omega\cdot\nabla)u$ behaves as if a hidden geometric constraint, an "invisible object", were shaping three-dimensional vorticity. The programme has two steps. (i) *Derive* what such an object can be from symmetry and energy considerations alone, in Hamiltonian/Lie–Poisson language. (ii) *Insert* the derived object into the dynamics, at vorticity level, through the Leray projection P.

Two pictures from the author's June 22, 2026 compilation guide the derivation. Both are formalized:

- **Chevalley–Eilenberg 2-cocycle.** Antisymmetric corrections to a Lie–Poisson bracket are CE 2-cochains. Jacobi requires them to be cocycles, and $d \circ d = 0$ makes coboundaries automatic cocycles. A *symmetric* correction is not a CE cochain at all. That is why the tether is metriplectic and not Hamiltonian (Prop. 2.7).
- **T^3 shear.** The shear $(\sin y, 0, 0)$ is divergence free, so it is admissible. The variant $(\sin x, 0, 0)$ is not (its divergence is 1 at the origin). This is the counterexample that shows the divergence-free constraint is essential (Prop. 2.8).

Whether real fluids carry the resulting term is *not* decided here (Section 4).

2 Result 1: geometric uniqueness of $\mathfrak{T}_F$

Throughout, H is a real inner-product space with $\dim H \geq 3$, $u \in H$, $u \neq 0$, and $\Pi_u x = x - \frac{\langle u,x\rangle}{\|u\|^2}u$.

Definition 2.1. $\mathfrak{T}_F[\kappa, w, u](F, G) := -(\kappa w)\langle \Pi_u F, \Pi_u G\rangle$.
Lean: `Frohman.Tether.frohmanianTether; frohmanianTether_apply`.

Lemma 2.2 (Basic properties)**.** *(a) $\mathfrak{T}_F$ is symmetric. (b) $\mathfrak{T}_F(F, F) \le 0$ whenever $\kappa w \ge 0$. (c) It is degenerate with respect to the kinetic-energy gradient: $\mathfrak{T}_F(F, u) = 0$ for all F. (d) It is invariant under every isometry commuting with Π_u.*

Proof. (a) is symmetry of the inner product. (b) gives $-\kappa w\|\Pi_u F\|^2$. (c) holds because $\Pi_u u = 0$. (d) If A is an isometry with $A\Pi_u = \Pi_u A$, then $\langle \Pi_u AF, \Pi_u AG\rangle = \langle A\Pi_u F, A\Pi_u G\rangle = \langle \Pi_u F, \Pi_u G\rangle$. □

Lean: `frohmanianTether_symm, frohmanianTether_self_nonpos, degeneracy_verification, frohmanianTether_invariant, piU_comm_isometry`.

Theorem 2.3 (Uniqueness up to scale)**.** *Let B be a symmetric bilinear form that is (C1) invariant under the coadjoint isometries fixing u and (C2) degenerate with respect to u. Then $B = \mathfrak{T}_F[c, 1, u]$ for some $c \in \mathbb{R}$. If, in addition, (C3) fixes the coefficient ($C3(\mathfrak{T}_F[c, 1, u]) \Rightarrow c = \kappa w$), then $B = \mathfrak{T}_F[\kappa, w, u]$.*

Proof. (C2) and symmetry give $B(x, y) = B(\Pi_u x, \Pi_u y)$ (`degeneracy_forces_projection`). So B is determined by its restriction to $u^\perp$, which has dimension ≥ 2. Fix a unit $e_0 \perp u$ (`exists_unit_orthogonal`). For every unit $e \perp u$, (C1) supplies an isometry fixing u that sends $e_0 \mapsto e$, so $B(e, e) = B(e_0, e_0)$ (`diag_eq_of_norm_eq`). By homogeneity, $B(x, x) = B(e_0, e_0)\|\Pi_u x\|^2$. Polarization recovers $B(x, y) = B(e_0, e_0)\langle \Pi_u x, \Pi_u y\rangle$, that is, $B = \mathfrak{T}_F[c, 1, u]$ with $c = -B(e_0, e_0)$. The calibrated statement then follows from (C3). □

Lean: `admissible_eq_smul_frohmanianTether, uniqueness_of_minimal_metric_tether, uniqueness_of_m`

Theorem 2.4 (Route A determines κ)**.** *Let $\omega \perp u$, $\|\omega\| = 1$, $w \ne 0$. Suppose $\mathfrak{T}_F[\kappa, w, u]$ matches the stretching coefficient S in the sense $\mathfrak{T}_F[\kappa, w, u](\omega, \omega) + Sw = 0$. Then $\kappa = S$. Consequently, in* $\dim \ge 3$ *with $S \ge 0$, $w > 0$, a correction is admissible for Route A iff $B = \mathfrak{T}_F[\kappa, w, u]$ with $\kappa = S$.*

Proof. Since $\omega \perp u$, $\Pi_u \omega = \omega$, so $\mathfrak{T}_F(\omega, \omega) = -\kappa w\|\omega\|^2 = -\kappa w$. The matching gives $(S - \kappa)w = 0$, and $w \ne 0$ yields $\kappa = S$. The equivalence combines this with Theorem 2.3. □

Lean: `routeA_determines_kappa, frohmanian_symplectic_tether_theorem, uniqueness_routeA_iff, frohmanianTether_routeA`.

Theorem 2.5 (Nine-term Jacobi)**.** *For a classical (Jacobi) bracket P and the tethered bracket $P + \mathfrak{T}_F$, the Jacobiator expands into nine terms. The three purely classical ones cancel by Jacobi of P, and the three purely tether terms vanish. So the Jacobiator equals the cyclic sum of the mixed terms, and Jacobi holds on a set R iff that cyclic sum vanishes on R. It vanishes when u is central for P, P is homogeneous in its first slot, and $\Pi_u P(a, b) = 0$.*

Proof. Bilinear expansion gives the nine terms (`jacobiator_nine_terms`). Terms 7–9 are $\mathfrak{T}_F$-in-$\mathfrak{T}_F$ and vanish because the tether vector field is Π_u-valued and annihilated in the second application (`tetherVec_tetherVec`, `jacobi_term7`–`jacobi_term9`). The classical triple cancels (`jacobi_classical_cancel`). Under the sufficient hypotheses each mixed term contains $\Pi_u P(\cdot, \cdot) = 0$ or $P(u, \cdot) = 0$. □

Lean: `jacobiator_eq_mixed, jacobi_iff_mixed_cancel, mixedTerm_eq_zero, jacobiator_eq_zero_of_ce`

Theorem 2.6 (Metriplectic split)**.** *Every bilinear form splits uniquely into symmetric and antisymmetric parts. $\mathfrak{T}_F$ is purely symmetric, and the metriplectic bracket $P + \mathfrak{T}_F$ with P antisymmetric conserves kinetic energy: $\{E, E\} = 0$ and $\mathfrak{T}_F(F, \nabla E) = 0$.*

Proof. $B = \frac{1}{2}(B + B^\top) + \frac{1}{2}(B - B^\top)$. Uniqueness holds because a form that is both symmetric and antisymmetric is zero. Energy conservation follows from $P(F, F) = 0$ and degeneracy. □

Lean: `split_unique, symPart_frohmanianTether, antiPart_frohmanianTether, metriplecticBracket_sp metriplectic_energy_conserved`.

Proposition 2.7 (Chevalley–Eilenberg)**.** *(a) $d_2 \circ d_1 = 0$ on 1-cochains of any Lie algebra. (b) Coboundaries are cocycles. (c) If 2 is invertible and the symmetric part of β is alternating, it is zero. So a nonzero symmetric correction such as $\mathfrak{T}_F$ is never a CE 2-cochain.*

Proof. (a) $d_2(d_1 f)(x, y, z)$ is $-f$ applied to the Jacobi sum $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$. (b) follows from (a). (c) $s(x, y) = s(y, x)$ and $s(x, y) = -s(y, x)$ give $2s = 0$. □

Lean: `d2_d1, cocycle_of_coboundary, antiCorr_cocycle_iff, symPart_not_cochain`.

Proposition 2.8 (T^3 shear)**.** $\operatorname{div}(\sin y, 0, 0) \equiv 0$*, whereas* $\operatorname{div}(\sin x, 0, 0) = \cos x$*, which equals* 1 *at the origin.*

Lean: `shear_divFree, shearBad_div, shearBad_not_divFree`.

3 Result 2: the tethered system

3.1 Insertion through the projection

The inserted system is

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega - \kappa P(|\omega|^2 \omega). \tag{1}$$

At velocity level it is the forced NS equation with force $F = \operatorname{curl}^{-1}(-\kappa P(|\omega|^2 \omega))$. **This is a modified Navier–Stokes system, not plain Navier–Stokes.** The quartic weight is built in, not ansatzed: $\mathfrak{T}_F[\kappa, \|\omega\|^2, u](\omega, \omega) = -\kappa \|\omega\|^4$ for $\omega \perp u$ (`quarticTether_eq`).

Theorem 3.1 (Curl of forced NS)**.** *If (u, p) is a classical smooth solution of $\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + F$ with $\operatorname{div} u = 0$, then $\omega = \operatorname{curl} u$ satisfies $\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega + \operatorname{curl} F$.*

Proof. Take the curl. Mixed partials commute on smooth fields, so $\operatorname{curl} \nabla p = 0$ and $\operatorname{curl} \Delta = \Delta \operatorname{curl}$, and ∂_t commutes with curl. The identity $\operatorname{curl}((u \cdot \nabla)u) = (u \cdot \nabla)\omega - (\omega \cdot \nabla)u$ for $\operatorname{div} u = 0$ is checked componentwise. □

Lean: `curlOfForcedNS, curl_grad, curl_laplacian, curl_advection, hasDerivAt_pd_time`.

Theorem 3.2 (Biot–Savart on trigonometric polynomials)**.** *For each nonzero wavevector k and amplitude $a \perp k$, the mode $b = \frac{ik \times a}{|k|^2} e^{ik \cdot x}$ satisfies $\operatorname{curl} b = a\, e^{ik \cdot x}$ and $\operatorname{div} b = 0$, and it is the unique such divergence-free mode. Summing gives a right inverse of curl on mean-zero divergence-free trigonometric polynomials.*

Proof. $\operatorname{curl} b = \frac{ik \times (ik \times a)}{|k|^2} e^{ik \cdot x} = \frac{-(k \cdot a)k + |k|^2 a}{|k|^2} e^{ik \cdot x} = a\, e^{ik \cdot x}$, and $\operatorname{div} b \propto k \cdot (k \times a) = 0$. Curl is linear over finite sums. □

Lean: `curlC_bsC, divC_bsC, bsC_unique, curl_bsMode, div_bsMode, curl_bsSum, curlRightInverse_tr`

Corollary 3.3 (Velocity form ⇒ vorticity form)**.** *If curl^{-1} is a right inverse of curl on the admissible class containing $-\kappa P(|\omega|^2 \omega)$, then every velocity solution of the tethered NS solves* (1)*.*

Lean: `tetheredVorticity_of_velocity, curl_lerayTetherForce`.

3.2 L^2 (enstrophy) bound

Lemma 3.4. *If P is a symmetric idempotent and $P\omega = \omega$, then $\langle \omega, -\kappa P g\rangle = -\kappa\langle \omega, g\rangle$. With $g = |\omega|^2\omega$ this equals $-\kappa\|\omega\|_4^4$.*

Lean: `inner_leray_of_divFree, tether_L2_term, inner_cubic`.

Theorem 3.5. *Let $E = \frac{1}{2}\|\omega\|_2^2$ and $Q = \|\omega\|_4^4$. Assume $E' = S - \nu D - \kappa Q$ (Cited Prop. 1), $S \le C\sqrt{2E}\sqrt{Q}$ (Cited Prop. 2), and $(2E)^2 \le Q$ (Cited Prop. 3). Then for $\kappa > 0$, $E(t) \le \max(E_0,\, C^2/(2\kappa^2))$ on $[0,T]$.*

Proof. Young with $\lambda = \kappa/2$: $S \le \frac{\kappa}{2}Q + \frac{C^2\cdot 2E}{2\kappa}$. Dropping $-\nu D \le 0$ and using $Q \ge (2E)^2$ gives the logistic inequality $E' \le \frac{C^2}{\kappa}E - 2\kappa E^2$ (`enstrophy_deriv_logistic`), so $E' < 0$ whenever $E > C^2/(2\kappa^2)$. A barrier argument (`image_le_of_deriv_right_lt_deriv_boundary'`) with a level M' slightly above the maximum, followed by letting $M' \downarrow \max$, gives the bound. □

Lean: `young_lam, stretch_absorb, enstrophy_deriv_le, enstrophy_deriv_logistic, tethered_enstrop tethered_L2_vorticity_bounded`.

3.3 H^1 (palinstrophy) estimate

Lemma 3.6 (Tether is sign-definite at H^1)**.** *For $\omega \in \mathbb{R}^3$ and $D = \nabla\omega$, $\sum_{ij} D_{ij}\,\partial_j(|\omega|^2\omega_i) = |\omega|^2|D|^2 + \frac{1}{2}\sum_j(2\sum_k \omega_k D_{kj})^2 \ge 0$, with no alignment assumption.*

Lean: `tether_H1_density_eq, tether_H1_density_nonneg, tether_H1_term`.

Theorem 3.7. *Let $G = \|\nabla\omega\|_2^2$, $L = \|\Delta\omega\|_2$. Assume $G' = 2(-\nu L^2 - \kappa J + N)$ with $J \ge 0$ (Cited Prop. 4) and $N \le A\,G^{1/4}L^{3/2}$ (Cited Prop. 5). Then $G(t) \le G(0)\exp\big(\frac{27A^4}{16\nu^3}t\big)$.*

Proof. Drop $-\kappa J \le 0$. Young with exponents $(4, 4/3)$: $AG^{1/4}L^{3/2} \le \nu L^2 + \frac{27A^4}{32\nu^3}G$ (`young_43`, `amgm_four`). So $G' \le \frac{27A^4}{16\nu^3}G$, and Grönwall (`le_gronwallBound_of_liminf_deriv_right_le`) finishes. □

Lean: `palin_deriv_le, tethered_palinstrophy_gronwall, higherNormControl_tethered`.

3.4 Log-loss comparison and global smoothness

At a spatial maximum of $|\omega|$, with $y = \|\omega\|_\infty$, the maximum principle gives a nonpositive viscous term (proved: `viscousAtMax_of_max`, `periodic_exists_max`). The tether gives $-\kappa y^3$ (proved from $\mathfrak{T}_F$: `tetherFeedback_of_quartic`). With Cited Props. 7, 8, 9 this gives $y' \le C(1+Z_0)\,y + C\log(e+Z)\,y^2 - (\kappa/2 - C_{abs})y^3$ (`tether_pointwise_logloss`). The cubic dominates, so y stays below the largest root of the comparison field (`comparison_bound_logloss`).

Theorem 3.8 (Global smoothness of the tethered system, conditional)**.** *Assume the tethered log inequality (Cited Prop. 10), higher-norm control, BKM continuation (Cited Prop. 11) and local continuation (Cited Prop. 12). Then the solution is smooth on $[0,T]$ for every T. Higher-norm control is itself discharged by Theorems 3.5 and 3.7 together with Cited Prop. 6.*

Proof. Let $\sigma = \sup\{T : \text{smooth on } [0,T]\}$, assuming this set is bounded. On $[0,\sigma)$ the comparison bound gives a uniform $\sup y$. BKM continuation then gives smoothness on $[0,\sigma]$, and local continuation extends it past σ, a contradiction. □

Lean: `global_regularity_logloss, global_regularity, lemma_3_1_uniform_bound_and_continuation, frohmanian_symplectic_tether_theorem_global`.

3.5 Remaining cited propositions (explicit hypotheses, never axioms)

Each item below is a `Prop` hypothesis in Lean. No `axiom` or `sorry` is used.

Cited Proposition 1 (`TetheredEnstrophyIdentity`)**.** Integrated enstrophy identity for (1). It follows from multiplying by ω, integrating by parts on T^3, and the lemma above.

Cited Proposition 2 (`StrainL2Bound`)**.** $\int \omega \cdot S\omega \le C\|\omega\|_2\|\omega\|_4^2$: Calderón–Zygmund / Biot–Savart, classical.

Cited Proposition 3 (`L4L2Holder`)**.** $\|\omega\|_2^4 \le \|\omega\|_4^4$ on the unit-volume torus (Hölder).

Cited Proposition 4 (`TetheredH1Identity`)**.** The H^1 energy identity (Majda–Bertozzi [4] §3.2 for the NS part).

Cited Proposition 5 (`NonlinH1Bound`)**.** Sobolev/Gagliardo–Nirenberg bound of the nonlinear term [2, 3].

Cited Proposition 6 (`H1ToHsBootstrap`)**.** H^1 bounds on $[0,T)$ imply H^s bounds for all s. For plain NS this is classical parabolic regularity [2, 3, 4]. **The tether-specific step (controlling $\kappa P(|\omega|^2\omega)$ in H^s, which is plausible since $H^2 \subset L^\infty$ in 3D) is not in the literature and remains to be proved.**

Cited Proposition 7 (`EnergyIdentity`, `StretchingByStrain`)**.** Pointwise evolution at the maximum, and stretching $\le \|S\|_\infty y^2$.

Cited Proposition 8 (`BKMLogInequality`)**.** $\|S\|_\infty \le C(1 + \|\omega\|_2 + \|\omega\|_\infty \log(e + \|\omega\|_{H^s}))$ [1]; see also Brezis–Gallouët and Kozono–Taniuchi [5].

Cited Proposition 9 (`CrossControlled`)**.** The cross term at the maximum is absorbed with constant C_{abs}.

Cited Proposition 10 (`NSTetherLogInequality`, `NSTetherEnstrophyInequality`)**.** The integrated (time-continuous) form of the pointwise inequality along the solution.

Cited Proposition 11 (`BKMContinuation`)**.** Beale–Kato–Majda [1]: bounded $\|\omega\|_\infty$ on $[0,T)$ implies continuation. Stated for plain NS; its extension to (1) is assumed.

Cited Proposition 12 (`LocalContinuation`)**.** Local existence for smooth periodic data (Fujita–Kato [6], Kato [7]).

Cited Proposition 13 (`CurlRightInverse` beyond trigonometric polynomials)**.** Proved on trigonometric polynomials. The extension to the smooth admissible class requires a density/continuity argument not yet formalized.

Search of the author's own prior work for $H^1 \to H^2$. The author's earlier repositories were searched. `NS_Millennium_Proof` (`Blueprint.md`, June 19, 2026, and `Modules/AnalyticEstimates.lean`, Aug. 26, 2026), `fefferman-clay-navier-stokes` (`HighSetDepletionAndBKM.lean`, Oct. 25, 2026) and `Clay-` (`latex/section10_gaps.tex`, Oct. 7, 2026) all invoke "BKM + parabolic regularity bootstrap" as a *cited classical black box* (in one case a `sorry`). None contains an independent proof of the $H^1 \to H^2$ step, let alone the tether-specific one. These are cited as the provenance of the plan, not as a proof.

4 Open question, stated plainly

1. **Plain Navier–Stokes is unchanged.** For a velocity field that solves plain NS, nothing above adds a damping term. The author's October notes say so in these words: "Because $B(F, H) = 0$ by Proposition [prop:degen], the tether does not add a damping term to the PDE (eq:vort) itself" (`NS_Millennium_Proof/main.tex`, section "Scaling Analysis and Open Analytic Questions); "A degeneracy $B(F, H) \equiv 0$ on a kinetic-energy Hamiltonian leaves the energy field classical and does not insert $-\kappa M^2$ into the strain" and "Equation (eq:missing) is not an identity of the unmodified vorticity equation" (`certs/CubicSourceGap.tex`, 7 Oct. 2026); "Derivation of the rate from the unmodified vorticity equation remains open" (`STATUS-2026-10-10.md`). (Bracketed labels inside quotations are the cited files' own equation references.) The mechanism is therefore the *degeneracy* $\Pi_u u = 0$, not a statement that the Leray projection annihilates a velocity-level cubic source; those notes make no such claim. Here the cubic term is inserted by hand at vorticity level through P, which is why the results concern (1) only. The Lean predicates `IsTrueNS` and `IsTetheredNS` coincide only at $\kappa = 0$ (`tetheredNS_zero_iff`), and at $\kappa = 0$ no feedback is realized (`not_tetherRealizesFeedback_zero`).

2. **The Clay Millennium problem remains open.** Nothing in this paper or repository claims otherwise.

3. **Do real fluids carry this term?** Answering this would need either a derivation of $-\kappa P(|\omega|^2\omega)$ from plain NS (none is known) or experimental/numerical evidence. Result 1 says that *if* such a symmetric, invariant, energy-degenerate correction exists, it must be $\mathfrak{T}_F$.

References

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